

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285451

Kind Code

A1

Publication Date

September 11, 2025

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Ambiguous Lane Detection Event Miner

Abstract

A computer system obtains a road image captured by a vehicle. The computer system determines whether the road image is ambiguous for lane marker classification. When the computer system determines that the road image is ambiguous for lane marker classification, the computer system generates a labeled road image and adds the labeled road image to a corpus of training data for training a model to generate an autonomous driving model. The computer system distributes the autonomous driving model to one or more vehicles. The autonomous driving model is configured to process road images captured by the one or more vehicles to facilitate at least partially autonomously driving the one or more vehicles.

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Family ID: 84810815

Appl. No.: 18/945382

Filed: November 12, 2024

Related U.S. Application Data

parent US continuation 18082535 20221215 parent-grant-document US 12142061 child US 18945382

parent US continuation 17850648 20220627 parent-grant-document US 11551459 child US 18082535

Publication Classification

Int. Cl.: G06V20/56 (20220101); B60W60/00 (20200101); G06V10/764 (20220101); G06V10/774 (20220101); G06V20/70 (20220101)

U.S. Cl.:

CPC **G06V20/588** (20220101); **G06V10/764** (20220101); **G06V10/774** (20220101);
G06V20/70 (20220101); B60W60/001 (20200201); B60W2420/403 (20130101);
B60W2552/53 (20200201)

Background/Summary

RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/082,535, filed Dec. 15, 2022, titled “Ambiguous Lane Detection Event Miner,” which is a continuation of U.S. patent application Ser. No. 17/850,648, filed Jun. 27, 2022, titled “Ambiguous Lane Detection Event Miner” now U.S. Pat. No. 11,551,459, issued Jan. 10, 2023, each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present application generally relates to vehicle technology, and more particularly to computer-aided methods and systems for at least partial autonomous vehicle control.

BACKGROUND

[0003] Modern vehicles are now capable of self-driving with different levels of autonomy. Each of these levels are characterized by the varying amount of human and/or autonomous control. For example, the Society of Automotive Engineers (SAE) defines 6 levels of driving automation ranging from 0 (fully manual) to 5 (fully autonomous). These levels have been adopted by the U.S. Department of Transportation.

[0004] Autonomous vehicles have numerous advantages, including lowering the number of vehicles on the roads (most privately owned vehicles are driven a small fraction of the time); promising more predictable and safer driving behavior than human driven vehicles; emitting less emissions if more vehicles are electrically powered; providing improved fuel efficiency; providing increased lane capacity; providing shorter travel times; allowing mobility for users who are incapable of driving; etc. One of the key obstacles facing the autonomous vehicle industry, however, is the complexity and unpredictability of road and traffic conditions. This makes it difficult to train autonomous vehicles for every possible rare condition or event that the vehicle may encounter while driving (so-called edge cases). For example, occasionally, human drivers may need to react to extraordinary or rare events, such as a package falling off a truck, a lane closure, or something even rarer, like an aircraft making an emergency landing on the freeway. In these situations, human drivers are often able to instinctively react to avoid harm to themselves and their vehicle, but unless the autonomous driving model has been trained for such a rare event, the vehicle may not know how to react.

SUMMARY

[0005] Currently, fleet operators often collect large amounts of data from individual vehicles in order to learn from existing road and traffic conditions. Typically, this data is sent from the vehicles to a remote server for storage and analysis (e.g., at a later time). Transmitting such large amounts of data (e.g., HD video or LIDAR data) from many vehicles (e.g., over a cellular data network) consumes valuable communication bandwidth and is prohibitively expensive. Furthermore, a lot of the data may be repetitive, typical, and do not represent rare events from which autonomous driving models can learn.

[0006] Accordingly, there is a need for improved systems, methods, and devices that provide a more efficient mechanism for collecting, monitoring, and learning from road condition data captured by a fleet of vehicles, such as data pertaining to lane markers (e.g., lane markings).

[0007] The present disclosure describes an improved system and method for lane marker detection

for vehicle driving. In accordance with some embodiments disclosed herein, a computer system (e.g., an event miner) determines a ratio of pixels having an ambiguous lane marker classification in a road image collected by a vehicle, identifies “interesting events” associated with lane marker detection in the road image, and determines whether the road image is an ambiguous image for lane marker classification. In accordance with a determination that the road image is an ambiguous image for lane marker classification, the computer system enables labeling of the image and adds the labeled image into a corpus of training images for retraining a model for autonomous driving. [0008] As disclosed herein, in some embodiments, the labeled image includes information identifying one or more lane markers (e.g., locations of the lane markers) in the respective labeled image. In some embodiments, the labeled image includes information identifying one or more lanes in the respective labeled image.

[0009] As disclosed herein, in some embodiments, retraining the model includes modifying an autonomous driving model based at least in part on the added labeled images (e.g., and the ambiguous image) such that the one or more vehicles can be driven at least partially autonomously using the autonomous driving model.

[0010] Accordingly, the systems and/or methods disclosed herein advantageously improve the autonomous driving models. For example, adding labeled image into an existing corpus of training images expands the training dataset. The labeled images, which include information identifying one or more lane markers, include ground truth of the respective ambiguous images. Retraining the model using the labeled images enables the retrained mode to recognize lane markers in images that would otherwise be ambiguous images, thereby improving performance of the model. Availability of the ground truth enables partially supervised learning to be performed. Furthermore, the event miner flags a subset, and not all, of targeted data corresponding to interesting events, and a reduced volume of targeted data is collected and stored, thereby leading to bandwidth and cost savings.

[0011] In accordance with some embodiments, a method for lane marker detection for vehicle driving is performed at a computer system. The computer system includes one or more processors and memory. The method includes obtaining a plurality of road images captured by one or more cameras attached to one or more vehicles. The one or more vehicles are configured to execute a model that facilitates driving of the one or more vehicles. The method further includes, for each road image of the plurality of road images, determining, in the road image, a fraction of pixels having an ambiguous lane marker classification. The method further includes based on the fraction of pixels, determining whether the road image is an ambiguous image for lane marker classification. The method further includes, in accordance with a determination that the road image is an ambiguous image for lane marker classification, enabling labeling of the image and adding the labeled image into a corpus of training images for retraining the model.

[0012] According to another aspect of the present application, a computer system includes one or more processors and memory. The memory stores one or more programs configured for execution by the one or more processors. The programs, when executed by the one or more processors, cause the computer system to perform the method for lane marker detection for vehicle driving as described above.

[0013] According to another aspect of the present application, a non-transitory computer readable storage medium stores a plurality of programs configured for execution by a computer system having one or more processors and memory, a plurality of sensors, and a vehicle control system (e.g., further including a plurality of actuators). The programs, when executed by the one or more processing units, cause the computer system to perform the method for lane marker detection for vehicle driving as described above.

[0014] Note that the various embodiments described above can be combined with any other embodiments described herein. The features and advantages described in the specification are not all inclusive and, in particular, many additional features and advantages will be apparent to one of

ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes and may not have been selected to delineate or circumscribe the inventive subject matter.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0015] The accompanying drawings, which are included to provide a further understanding of the embodiments and are incorporated herein and constitute a part of the specification, illustrate the described embodiments and together with the description serve to explain the underlying principles.

[0016] FIG. 1 is an example vehicle driving environment for a plurality of vehicles, in accordance with some embodiments.

[0017] FIG. 2 is a block diagram of an example vehicle configured to be driven with a certain level of autonomy, in accordance with some embodiments.

[0018] FIG. 3 is a block diagram of an example server for monitoring and managing vehicles in a vehicle driving environment, in accordance with some embodiments.

[0019] FIG. 4 is a block diagram of a machine learning system for training and applying vehicle data processing models for facilitating at least partial autonomous driving of a vehicle, in accordance with some embodiments.

[0020] FIG. 5A is a structural diagram of an example neural network applied to process vehicle data in a vehicle data processing model, in accordance with some embodiments, and FIG. 5B is an example node in the neural network, in accordance with some embodiments.

[0021] FIG. 6A is a flow diagram of an exemplary process of classifying an image on both a pixel level and an image level, in accordance with some embodiments.

[0022] FIG. 6B is a flow diagram of an exemplary process of processing an ambiguous image, in accordance with some embodiments.

[0023] FIGS. 7A to 7F are exemplary road images that have been identified as ambiguous images for lane marker classification, in accordance with some embodiments.

[0024] FIGS. 8A to 8C is a flow diagram of a method for lane marker detection for vehicle driving, in accordance with some embodiments.

[0025] Like reference numerals refer to corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

[0026] Reference will now be made in detail to specific embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous non-limiting specific details are set forth in order to assist in understanding the subject matter presented herein. But it will be apparent to one of ordinary skill in the art that various alternatives may be used without departing from the scope of claims and the subject matter may be practiced without these specific details. For example, it will be apparent to one of ordinary skill in the art that the subject matter presented herein can be implemented on many types of electronic devices with digital video capabilities.

[0027] FIG. 1 is an example vehicle driving environment **100** having a plurality of vehicles **102**, in accordance with some embodiments. Each vehicle **102** has one or more processors, memory, a plurality of sensors, and a vehicle control system, and is configured to sense the vehicle driving environment **100** and drive on roads having different road conditions. The plurality of vehicles **102** may be, for example, passenger cars **102P** (e.g., sport-utility vehicles and sedans), vans **102V**, trucks **102T**, and driver-less cars. Each vehicle **102** can collect sensor data or user inputs, execute

user applications, present outputs on its user interface, and/or operate the vehicle control system to drive the vehicle **102**. The collected data or user inputs can be processed locally (e.g., for training and/or for prediction) at the vehicle **102** and/or remotely by server(s) **104**. The one or more servers **104** provide system data (e.g., boot files, operating system images, and user applications) to the vehicle **102**, and in some embodiments, process the data and user inputs received from the vehicle **102** when the user applications are executed on the vehicle **102**. In some embodiments, the vehicle driving environment **100** further includes storage **106** for storing data related to the vehicles **102**, servers **104**, and applications executed on the vehicles **102**.

[0028] For each vehicle **102**, the plurality of sensors includes one or more of: a global positioning system (GPS) sensor, a light detection and ranging (LiDAR) scanner, one or more cameras, a radio detection and ranging (RADAR) sensor, an infrared sensor, one or more ultrasonic sensors, a dedicated short-range communication (DSRC) module, an inertial navigation system (INS) including accelerometers and gyroscopes, and/or an odometry sensor. The cameras are configured to capture a plurality of images in the vehicle driving environment **100**, and the plurality of images are applied to map the vehicle driving environment **100** to a 3D vehicle space and identify a location of the vehicle **102** within the environment **100**. The cameras also operate with one or more other sensors (e.g., GPS, LiDAR, RADAR, INS) to localize the vehicle **102** in the 3D vehicle space. For example, the GPS identifies a geographical position (geolocation) of the vehicle **102** on the Earth, and the INS measures relative vehicle speeds and accelerations between the vehicle **102** and adjacent vehicles **102**. The LiDAR scanner measures a distance of the vehicle **102** to adjacent vehicles **102** and other objects. Data collected by these sensors are used to determine vehicle locations determined from the plurality of images or facilitate determining vehicle locations between two images.

[0029] The vehicle control system includes a plurality of actuators for at least steering, braking, controlling the throttle (e.g., accelerating, maintaining a constant velocity, or decelerating), and transmission control. Depending on the level of automation, each of the plurality of actuators (or manually controlling the vehicle, e.g., by turning the steering wheel) can be controlled manually by a driver of the vehicle, automatically by the one or more processors of the vehicle, or jointly by both of the driver and processor(s). When the vehicle **102** controls the plurality of actuators independently or jointly with the driver, the vehicle **102** obtains the sensor data collected by the plurality of sensors, identifies adjacent road features in the vehicle driving environment **100**, tracks the motion of the vehicle, tracks the relative distance between the vehicle and any surrounding vehicles or other objects, and generates vehicle control instructions to at least partially autonomously control driving of the vehicle **102**. Conversely, in some embodiments, when the driver takes control of the vehicle, the driver manually provides vehicle control instructions via a steering wheel, a braking pedal, a throttle pedal, and a gear lever directly. In some embodiments, a vehicle user application is executed on the vehicle and configured to provide a user interface. The driver provides vehicle control instructions to control the plurality of actuators of the vehicle control system via the user interface of the vehicle user application. By these means, the vehicle **102** is configured to drive with its own vehicle control system and/or the driver of the vehicle **102** according to a certain level of autonomy.

[0030] In some embodiments, autonomous vehicles can include, for example, a fully autonomous vehicle, a partially autonomous vehicle, a vehicle with driver assistance, or an autonomous capable vehicle. The capabilities of autonomous vehicles can be associated with a classification system or taxonomy having tiered levels of autonomy. A classification system can be specified by, for example, industry standards or governmental guidelines. For example, the levels of autonomy can be considered using a taxonomy such as level 0 (momentary driver assistance), level 1 (driver assistance), level 2 (additional assistance), level 3 (conditional assistance), level 4 (high automation), and level 5 (full automation without any driver intervention). Following this example, an autonomous vehicle can be capable of operating, in some instances, in at least one of levels 0

through 5. According to various embodiments, an autonomous capable vehicle may refer to a vehicle that can be operated by a driver manually (that is, without the autonomous capability activated) while being capable of operating in at least one of levels 0 through 5 upon activation of an autonomous mode. As used herein, the term “driver” may refer to a local operator (e.g., an operator in the vehicle) or a remote operator (e.g., an operator physically remote from and not in the vehicle). The autonomous vehicle may operate solely at a given level (e.g., level 2 additional assistance or level 5 full automation) for at least a period of time or during the entire operating time of the autonomous vehicle. Other classification systems can provide other levels of autonomy characterized by different vehicle capabilities.

[0031] More specifically, in some embodiments, the vehicle **102** drives in the vehicle driving environment **100** at level 5. The vehicle **102** collects sensor data from the plurality of sensors, processes the sensor data to generate vehicle control instructions, and controls the vehicle control system to drive the vehicle autonomously in response to the vehicle control instructions.

Alternatively, in some situations, a vehicle **102** drives in the vehicle driving environment **100** at level 0. The vehicle **102** collects the sensor data and processes the sensor data to provide feedback (e.g., a warning or alert) to a driver of the vehicle **102** to allow the driver to drive the vehicle **102** manually and based on the driver's own judgement. Alternatively, in some situations, a vehicle **102** drives in the vehicle driving environment **100** partially autonomously at one of levels 1-4. The vehicle **102** collects the sensor data and processes the sensor data to generate a vehicle control instruction for a portion of the vehicle control system and/or provide feedback to a driver of the vehicle **102**. The vehicle **102** is driven jointly by the vehicle control system of the vehicle **102** and the driver of the vehicle **102**. Optionally, the vehicle control system and driver of the vehicle **102** control different portions of the vehicle **102**. Optionally, the vehicle **102** determines a vehicle status. Based on the vehicle status, a vehicle control instruction of one of the vehicle control system or driver of the vehicle **102** preempts or overrides another vehicle control instruction provided by the other one of the vehicle control system or driver of the vehicle **102**.

[0032] For vehicle **102**, the sensor data collected by the plurality of sensors, the vehicle control instructions applied to the vehicle control system, and the user inputs received via the vehicle user application form a collection of vehicle data **112**. In some embodiments, at least a subset of vehicle data **112** from each vehicle **102** is provided to one or more servers **104**. A server **104** provides a central vehicle platform for collecting and analyzing the vehicle data **112**, monitoring vehicle operation, detecting faults, providing driving solutions, and updating additional vehicle information **114** to individual vehicles **102** or client devices **108**. In some embodiments, the server **104** manages vehicle data **112** of each individual vehicle **102** separately. In some embodiments, the server **104** consolidates vehicle data **112** from multiple vehicles **102** and manages the consolidated vehicle data jointly, e.g., statistically aggregates the data.

[0033] Additionally, in some embodiments, the vehicle driving environment **100** further includes one or more client devices **108**, e.g., desktop computers, laptop computers, tablet computers, and mobile phones. Each client device **108** is configured to execute a client user application associated with the central vehicle platform provided by the server **104**. The client device **108** is logged onto a user account on the client user application, and the user account is associated with one or more vehicles **102**. The server **104** provides the collected vehicle data **112** and additional vehicle information **114** (e.g., vehicle operation information, fault information, or driving solution information) for the one or more associated vehicles **102** to the client device **108** using the user account of the client user application. In some embodiments, the client device **108** is located in the one or more vehicles **102**, while in other embodiments, the client device is at a location distinct from the one or more associated vehicles **102**. As such, the server **104** can apply its computational capability to manage the vehicle data and facilitate vehicle monitoring and control on different levels, e.g., for each individual vehicle, for a collection of vehicles, and for related client devices **108**.

[0034] The plurality of vehicles **102**, one or more servers **104**, and one or more client devices **108** are communicatively coupled to each other via one or more communication networks **110**, which are used to provide communications links between these vehicles and computers connected together within the vehicle driving environment **100**. The one or more communication networks **110** may include connections, such as a wired network, wireless communication links, or fiber optic cables. Examples of the one or more communication networks **110** include local area networks (LAN), wide area networks (WAN) such as the Internet, or a combination thereof. The one or more communication networks **110** are, optionally, implemented using any known network protocol, including various wired or wireless protocols, such as Ethernet, Universal Serial Bus (USB), FIREWIRE, Long Term Evolution (LTE), Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wi-Fi, voice over Internet Protocol (VoIP), Wi-MAX, or any other suitable communication protocol. A connection to the one or more communication networks **110** may be established either directly (e.g., using 3G/4G connectivity to a wireless carrier), or through a network interface (e.g., a router, switch, gateway, hub, or an intelligent, dedicated whole-home control node), or through any combination thereof. In some embodiments, the one or more communication networks **110** allow for communication using any suitable protocols, like Transmission Control Protocol/Internet Protocol (TCP/IP). In some embodiments, each vehicle **102** is communicatively coupled to the server(s) **104** via a cellular communication network.

[0035] In some embodiments, deep learning techniques are applied by the vehicles **102**, server(s) **104**, or both to process the vehicle data **112**. For example, in some embodiments, after image data are collected by the cameras of one of the vehicles **102**, the image data is processed using an object detection model to identify objects (e.g., road features including, but not limited to, vehicles, lane lines, lane markers (e.g., lane markings), shoulder lines, road dividers, traffic lights, traffic signs, road signs, cones, a pedestrian, a bicycle, and a driver of the first vehicle) in the vehicle driving environment **100**. In some embodiments, additional sensor data are also collected and processed by a vehicle control model to generate a vehicle control instruction for controlling the vehicle control system. In some embodiments, a vehicle planning model may also be applied to plan a driving control process based on the collected sensor data and the vehicle driving environment **100**. The object detection model, vehicle control model, and vehicle planning model are collectively referred to herein as vehicle data processing models, each of which includes one or more neural networks. Such a vehicle data processing model is optionally applied by the vehicles **102**, server(s) **104**, or both to process the vehicle data **112** to infer associated vehicle status and/or provide control signals. In some embodiments, a vehicle data processing model is trained by a server **104**, and applied locally or provided to one or more vehicles **102** for inference of the associated vehicle status and/or to provide control signals. Alternatively, a vehicle data processing model is trained locally by a vehicle **102**, and applied locally or shared with one or more other vehicles **102**, e.g., by way of the server **104**. A vehicle data processing model is optionally trained in a supervised, semi-supervised, or unsupervised manner.

[0036] FIG. 2 is a block diagram of an example vehicle **102** configured to be driven with a certain level of autonomy, in accordance with some embodiments. The vehicle **102** typically includes one or more processing units (CPUs) **202**, one or more network interfaces **204**, memory **206**, and one or more communication buses **208** for interconnecting these components (sometimes called a chipset). The vehicle **102** includes one or more user interface devices. The user interface devices include one or more input devices **210**, which facilitate user input, such as a keyboard, a mouse, a voice-command input unit or microphone, a touch screen display, a touch-sensitive input pad, a gesture capturing camera, or other input buttons or controls. Furthermore, in some embodiments, the vehicle **102** uses a microphone and voice recognition or a camera and gesture recognition to supplement or replace the keyboard. In some embodiments, the one or more input devices **210**

include one or more cameras, scanners, or photo sensor units for capturing images, for example, of a driver and a passenger in the vehicle **102**. The vehicle **102** also includes one or more output devices **212**, which enable presentation of user interfaces and display content, including one or more speakers and/or one or more visual displays (e.g., a display panel located near to a driver's right hand in right-hand-side operated vehicles typical in the U.S.).

[0037] The vehicle **102** includes a plurality of sensors **260** configured to collect sensor data in a vehicle driving environment **100**. The plurality of sensors **260** include one or more of a GPS **262**, a LiDAR scanner **264**, one or more cameras **266**, a RADAR sensor **268**, an infrared sensor **270**, one or more ultrasonic sensors **272**, a DSRC module **274**, an INS **276** including accelerometers and gyroscopes, and an odometry sensor **278**. The GPS **262** localizes the vehicle **102** in an Earth coordinate, e.g., using a latitude value and a longitude value, and can reach a first accuracy level up to less than 1 meter (e.g., 30 cm). The LiDAR scanner **264** uses light beams to estimate relative distances between the scanner **264** and a target object (e.g., another vehicle **102**), and can reach a second accuracy level better than the first accuracy level of the GPS **262**. The cameras **266** are installed at different locations of the vehicle **102** to monitor surroundings of the camera **266** from different perspectives. In some situations, a camera **266** is installed facing an interior of the vehicle **102** and configured to monitor a state of a driver of the vehicle **102**. The RADAR sensor **268** emits electromagnetic waves and collects reflected waves to determine a speed and a distance of an object over which the waves are reflected. The infrared sensor **270** identifies and tracks objects in an infrared domain when lighting conditions are poor. The one or more ultrasonic sensors **272** are used to detect objects in a short distance, e.g., to assist parking. The DSRC module **274** is used to exchange information with a road feature (e.g., a traffic light). The INS **276** uses the accelerometers and gyroscopes to measure a position, orientation, and speed of the vehicle. The odometry sensor **278** tracks a distance the vehicle **102** has travelled, e.g., based on a wheel speed. In some embodiments, based on the sensor data collected by the plurality of sensors **260**, the one or more processors **202** of the vehicle monitors its own vehicle state **282**, a driver or passenger state **284**, states of adjacent vehicles **286**, and road conditions **288** associated with a plurality of road features. [0038] Additionally, the vehicle **102** has a control system **290** including a steering control **292**, a braking control **294**, a throttle control **296**, a transmission control **298**, signaling and lighting controls, and other controls. In some embodiments, one or more actuators of the vehicle control system **290** are automatically controlled based on the plurality of sensor data collected by the plurality of sensors **260**, e.g., according to one or more of the vehicle state **282**, driver or passenger state **284**, states of adjacent vehicles **286**, and road conditions **288**.

[0039] The memory **206** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices. In some embodiments, the memory includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. In some embodiments, the memory **206** includes one or more storage devices remotely located from one or more processing units **202**. The memory **206**, or alternatively the non-volatile the memory within the memory **206**, includes a non-transitory computer readable storage medium. In some embodiments, the memory **206**, or the non-transitory computer readable storage medium of memory **206**, stores the following programs, modules, and data structures, or a subset or superset thereof: [0040] an operating system **214**, which includes procedures for handling various basic system services and for performing hardware dependent tasks; [0041] a network communication module **216**, which connects each vehicle **102** to other devices (e.g., another vehicle **102**, server **104**, client device **108**) via one or more network interfaces (wired or wireless) and one or more communication networks **110**, such as the Internet, other wide area networks, local area networks, metropolitan area networks, and so on; [0042] a user interface module **218**, which enables presentation of information (e.g., a graphical user interface for application **224**, widgets, websites and web pages thereof, audio content, and/or video content) at

the vehicle **102** via one or more output devices **212** (e.g., displays or speakers); [0043] an input processing module **220**, which detects one or more user inputs or interactions from one of the one or more input devices **210** and interprets the detected input or interaction; [0044] a web browser module **222**, which navigates, requests (e.g., via HTTP), and displays websites and web pages thereof, including a web interface for logging into a user account of a user application **224** associated with the vehicle **102** or another vehicle; [0045] one or more user applications **224**, which is executed at the vehicle **102** and includes a vehicle user application that controls the vehicle **102** and edits and reviews settings and data associated with the vehicle **102**; [0046] a model training module **226**, which trains a vehicle data processing model **250** that includes at least one neural network and is applied to process vehicle data (e.g., sensor data and vehicle control data) of the vehicle **102**; [0047] a data processing module **228**, which performs a plurality of on-vehicle tasks including, but not limited to, perception and object analysis **230**, vehicle localization and environment mapping **232**, vehicle drive control **234**, vehicle drive planning **236**, local operation monitoring **238**, and vehicle driving behavior monitoring **240**; [0048] Vehicle device database **242** that stores vehicle data **112**, including: [0049] device settings **243**, including common device settings (e.g., service tier, device model, storage capacity, processing capabilities, communication capabilities, and/or medical procedure settings) of the vehicle **102**; [0050] user account information **244** for the one or more user applications **224** (e.g., user names, security questions, account history data, user preferences, and predefined account settings); [0051] network parameters **246** for the one or more communication networks **110**, e.g., IP address, subnet mask, default gateway, DNS server and host name; [0052] training data **248** for training a vehicle data processing model **250**; [0053] vehicle data processing models **250** for processing vehicle data **112**, where in some embodiments, the vehicle data processing models **250** include an autonomous driving model **252** configured to drive the vehicle **102** at least partially autonomously; [0054] sensor data **254** captured or measured by the plurality of sensors **260**; [0055] mapping and location data **256** that are determined from the sensor data **254** to map the vehicle driving environment **100** and locations of the vehicle **102** in the environment **100**; and [0056] vehicle control data **258** that are automatically generated by the vehicle **102** or manually inputted by the user via the vehicle control system **290** to drive the vehicle **102**.

[0057] Each of the above identified elements may be stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, modules or data structures, and thus various subsets of these modules may be combined or otherwise re-arranged in various embodiments. In some embodiments, the memory **206** stores a subset of the modules and data structures identified above. In some embodiments, the memory **206** stores additional modules and data structures not described above. In some embodiments, a subset of the programs, modules, and/or data stored in the memory **206** are stored on and/or executed by the server system **104**.

[0058] FIG. **3** is a block diagram of an example server **104** for monitoring and managing vehicles **102** in a vehicle driving environment (e.g., the environment **100** in FIG. **1**), in accordance with some embodiments. Examples of the server **104** include, but are not limited to, a server computer, a desktop computer, a laptop computer, a tablet computer, or a mobile phone. The server **104** typically includes one or more processing units (CPUs) **302**, one or more network interfaces **304**, memory **306**, and one or more communication buses **308** for interconnecting these components (sometimes called a chipset). The server **104** includes one or more user interface devices. The user interface devices include one or more input devices **310**, which facilitate user input, such as a keyboard, a mouse, a voice-command input unit or microphone, a touch screen display, a touch-sensitive input pad, a gesture capturing camera, or other input buttons or controls. Furthermore, in some embodiments, the server **104** uses a microphone and voice recognition or a camera and gesture recognition to supplement or replace the keyboard. In some embodiments, the one or more

input devices **310** include one or more cameras, scanners, or photo sensor units for capturing images, for example, of graphic serial codes printed on electronic devices. The server **104** also includes one or more output devices **312**, which enable presentation of user interfaces and display content, including one or more speakers and/or one or more visual displays.

[0059] The memory **306** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices. In some embodiments, the memory includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. In some embodiments, the memory **306** includes one or more storage devices remotely located from one or more processing units **302**. The memory **306**, or alternatively the non-volatile memory within memory **306**, includes a non-transitory computer readable storage medium. In some embodiments, the memory **306**, or the non-transitory computer readable storage medium of the memory **306**, stores the following programs, modules, and data structures, or a subset or superset thereof: [0060] an operating system **314**, which includes procedures for handling various basic system services and for performing hardware dependent tasks; [0061] a network communication module **316**, which connects the server **104** to other devices (e.g., vehicles **102**, another server **104**, client devices **108**) via one or more network interfaces (wired or wireless) and one or more communication networks **110**, such as the Internet, other wide area networks, local area networks, metropolitan area networks, and so on; [0062] a user interface module **318**, which enables presentation of information (e.g., a graphical user interface for user application **324**, widgets, websites and web pages thereof, audio content, and/or video content) at the vehicle **102** via one or more output devices **312** (e.g., displays or speakers); [0063] an input processing module **320**, which detects one or more user inputs or interactions from one of the one or more input devices **310** and interprets the detected input or interaction; [0064] a web browser module **322**, which navigates, requests (e.g., via HTTP), and displays websites and web pages thereof, including a web interface for logging into a user account of a user application **324**; [0065] one or more user applications **324**, which is executed at the server **104** and includes a vehicle user application that associates vehicles **102** with user accounts and facilitates controlling the vehicles **102** and editing and reviewing settings and data associated with the vehicles **102**; [0066] a model training module **226**, which trains a vehicle data processing model **250** that includes at least one neural network and is applied to process vehicle data (e.g., sensor data and vehicle control data) of one or more vehicles **102**; [0067] a data processing module **228**, which manages a multi-vehicle operation monitoring platform **332** configured to collect vehicle data **112** from a plurality of vehicles **102**, monitor vehicle operation, detect faults, provide driving solutions, and update additional vehicle information **114** to individual vehicles **102** or client devices **108**, where the data processing module **228** optionally manages vehicle data **112** for each individual vehicle **102** separately or processes vehicle data **112** of multiple vehicles **102** jointly, e.g., statistically; [0068] vehicle server data **340**, including: [0069] device settings **342**, which include common device settings (e.g., service tier, device model, storage capacity, processing capabilities, communication capabilities, and/or medical procedure settings) of the server **104**; [0070] user account information **344** for the one or more user applications **324** (e.g., user names, security questions, account history data, user preferences, and predefined account settings); [0071] network parameters **346** for the one or more communication networks **110**, e.g., IP address, subnet mask, default gateway, DNS server and host name; [0072] training data **248** for training a vehicle data processing model **250**; [0073] vehicle data processing models **250** for processing vehicle data, where the vehicle data processing models **250** include an autonomous driving model **252** configured to drive the vehicle **102** at least partially autonomously; [0074] vehicle data **112**, which is collected from a plurality of vehicles **102** and includes sensor data **254**, mapping and location data **256**, and vehicle control data **258**; and [0075] additional vehicle information **114** including vehicle operation information, fault information, or driving solution information, which are generated from the collected vehicle data

112.

[0076] In some embodiments, the model training module **226** includes a training data augmentation module **328** configured to synthesize training data based on a predefined dataset or collected sensor data (or camera data) of the vehicles **102**. Optionally, the predefined dataset is used with the synthesized training data to train a vehicle data processing model **250**. Optionally, the collected sensor data is optionally used with the synthesized training data to train a vehicle data processing model **250**. Optionally, the synthesized training data is used independently to train a vehicle data processing model **250**. By these means, the training data can be augmented conveniently, allowing the vehicle data processing model **250** to be trained efficiently and offer a higher accuracy level.

[0077] Each of the above identified elements may be stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, modules or data structures, and thus various subsets of these modules may be combined or otherwise re-arranged in various embodiments. In some embodiments, the memory **306** stores a subset of the modules and data structures identified above. In some embodiments, the memory **306** stores additional modules and data structures not described above. In some embodiments, a subset of the programs, modules, and/or data stored in the memory **306** are stored on and/or executed by the vehicle **102**.

[0078] FIGS. **4**, **5A**, and **5B** provide background on the machine learning systems described herein, which are helpful in understanding the details of the embodiments described from FIG. **6A** onward.

[0079] FIG. **4** is a block diagram of a machine learning system **400** for training and applying vehicle data processing models **250** for facilitating driving of a vehicle, in accordance with some embodiments. The machine learning system **400** includes a model training module **226** establishing one or more vehicle data processing models **250** and a data processing module **228** for processing vehicle data **112** using the vehicle data processing model **250**. In some embodiments, both of the model training module **226** (e.g., the model training module **226** in FIG. **2**) and the data processing module **228** are located within the vehicle **102**, while a training data source **404** provides training data **248** to the vehicle **102**. The training data source **404** is optionally the data obtained from the vehicle **102** itself, a server **104**, storage **106**, or another vehicle or vehicles **102**. Alternatively, in some embodiments, the model training module **226** (e.g., the model training module **226** in FIG. **3**) is located at a server **104**, and the data processing model(s) **228** are located in a vehicle **102**. The server **104** trains the data processing models **250** and provides the trained models **250** to the vehicle **102** to process real-time vehicle data **112** detected by the vehicle **102**. In some embodiments, the training data **248** provided by the training data source **404** include a standard dataset (e.g., a set of road images) widely used by engineers in the autonomous vehicle industry to train vehicle data processing models **250**. In some embodiments, the training data **248** include vehicle data **112** and/or additional vehicle information **114**, which are collected from one or more vehicles **102** that will apply the vehicle data processing models **250** or from distinct vehicles **102** that will not apply the vehicle data processing models **250**. The vehicle data **112** further includes one or more of sensor data **254**, road mapping and location data **256**, and control data **258**. Further, in some embodiments, a subset of training data **248** is modified to augment the training data **248**. The subset of modified training data is used in place of or jointly with the subset of training data **248** to train the vehicle data processing models **250**.

[0080] In some embodiments, the model training module **226** (FIG. **2**) includes a model training engine **410**, and a loss control module **412**. Each vehicle data processing model **250** is trained by the model training engine **410** to process corresponding vehicle data **112** to implement a respective one of a plurality of on-vehicle tasks including, but not limited to, perception and object analysis **230**, vehicle localization and environment mapping **232**, vehicle drive control **234**, vehicle drive planning **236**, local operation monitoring **238**, and vehicle driving behavior monitoring **240**. Specifically, the model training engine **410** receives the training data **248** corresponding to a

vehicle data processing model **250** to be trained, processes the training data using the vehicle data processing model **250**, and generates an output from each training data item. In some embodiments, during this process, the loss control module **412** monitors a loss function comparing the output associated with the respective training data item to a ground truth of the respective training data item. In this embodiment, the model training engine **410** modifies the vehicle data processing model(s) **250** to reduce any loss function, until the loss function satisfies a loss criteria (e.g., a comparison result of the loss function is minimized or reduced below a loss threshold). The vehicle data processing model(s) **250** are thereby trained and provided to the data processing module **228** of a vehicle **102** to process real-time vehicle data **112** from the vehicle.

[0081] In some embodiments, the model training module **226** further includes a data pre-processing module **408** configured to pre-process the training data **248** before the training data **248** is used by the model training engine **410** to train a vehicle data processing model **250**. For example, an image pre-processing module **408** is configured to process road images in the training data **248** to a predefined image format, e.g., normalize the road images to a fixed size, resolution or contrast level. In another example, an image pre-processing module **408** extracts a region of interest (ROI) corresponding to a drivable area in each road image or separate content of the drivable area into a distinct image.

[0082] In some embodiments, the model training module **226** offers supervised learning in which the training data **248** is labelled and includes a desired output for each training data item (also called the ground truth in some situations). The desirable output is optionally labelled manually by human or automatically by the model training model **226** before training. Conversely, in some embodiments, the model training module **226** offers unsupervised learning in which the training data **248** are not labelled. The model training module **226** is configured to identify previously undetected patterns in the training data **248** without pre-existing labels and with no or little human supervision. Additionally, in some embodiments, the model training module **226** offers partially supervised learning in which the training data are partially labelled.

[0083] In some embodiments, the data processing module **228** includes a data pre-processing module **414**, a model-based processing module **416**, and a data post-processing module **418**. The data pre-processing modules **414** pre-processes vehicle data **112** based on the type of the vehicle data **112**. In some embodiments, functions of the data pre-processing modules **414** are consistent with those of the pre-processing modules **408**, and convert the vehicle data **112** to a predefined data format that is suitable for the inputs of the model-based processing module **416**. The model-based processing module **416** applies the trained vehicle data processing model **250** provided by the model training module **226** to process the pre-processed vehicle data **112**. In some embodiments, the model-based processing module **416** can also monitor an error indicator to determine whether the vehicle data **112** has been properly processed in the vehicle data processing model **250**. In some embodiments, the processed vehicle data is further processed by the data post-processing module **418** in a preferred format or to provide additional vehicle information **114** that can be derived from the processed vehicle data. The data processing module **228** used the processed vehicle data to at least partially autonomously drive the vehicle **102**, e.g., at least partially autonomously. For example, the processed vehicle data includes vehicle control instructions that are used by the vehicle control system **290** to drive the vehicle **102**.

[0084] FIG. 5A is a structural diagram of an example neural network **500** applied to process vehicle data in a vehicle data processing model **250**, in accordance with some embodiments, and FIG. 5B is an example node **520** in the neural network **500**, in accordance with some embodiments. It should be noted that this description is used as an example only, and other types or configurations may be used to implement the embodiments described herein. The vehicle data processing model **250** is established based on the neural network **500**. A corresponding model-based processing module **416** applies the vehicle data processing model **250** including the neural network **500** to process vehicle data **112** that has been converted to a predefined data format. The neural network **500** includes a

collection of nodes **520** that are connected by links **512**. Each node **520** receives one or more node inputs **522** and applies a propagation function **530** to generate a node output **524** from the one or more node inputs. As the node output **524** is provided via one or more links **512** to one or more other nodes **520**, a weight w associated with each link **512** is applied to the node output **524**. Likewise, the one or more node inputs **522** are combined based on corresponding weights w_1 , w_2 , w_3 , and w_4 according to the propagation function **530**. In an example, the propagation function **530** is a product of a non-linear activation function **532** and a linear weighted combination of the one or more node inputs **522**.

[0085] The collection of nodes **520** is organized into one or more layers in the neural network **500**. Optionally, the one or more layers includes a single layer acting as both an input layer and an output layer. Optionally, the one or more layers includes an input layer **502** for receiving inputs, an output layer **506** for providing outputs, and zero or more hidden layers **504** (e.g., **504A** and **504B**) between the input layer **502** and the output layer **506**. A deep neural network has more than one hidden layer **504** between the input layer **502** and the output layer **506**. In the neural network **500**, each layer is only connected with its immediately preceding and/or immediately following layer. In some embodiments, a layer **502** or **504B** is a fully connected layer because each node **520** in the layer **502** or **504B** is connected to every node **520** in its immediately following layer **504A** or **506**, respectively. In some embodiments, one of the one or more hidden layers **504** includes two or more nodes that are connected to the same node in its immediately following layer for down sampling or pooling the two or more nodes **520**. Particularly, max pooling uses a maximum value of the two or more nodes in the layer **504B** for generating the node of the immediately following layer **506** (i.e., the output layer **506**) connected to the two or more nodes.

[0086] In some embodiments, a convolutional neural network (CNN) is applied in a vehicle data processing model **250** to process vehicle data (particularly, video and image data captured by cameras **266** of a vehicle **102**). The CNN employs convolution operations and belongs to a class of deep neural networks. The one or more hidden layers **504** of the CNN are convolutional layers convolving with a multiplication or dot product. Each node in a convolutional layer receives inputs from a receptive area associated with a previous layer (e.g., five nodes), and the receptive area is smaller than the entire previous layer and may vary based on a location of the convolution layer in the CNN. For the input layer **502**, a kernel is used to combine pixels in a respective area including each pixel to generate layer outputs (i.e., feature elements of a feature map). In an example, the kernel corresponds to a 3×3 matrix including weights applied to combine the pixels in the respective area of each pixel in a linear weighted manner. Video or image data is pre-processed to a predefined video/image format corresponding to the inputs of the CNN. The pre-processed video or image data is abstracted by each layer of the CNN to a respective feature map. By these means, video and image data can be processed by the CNN for video and image recognition or object detection.

[0087] In some embodiments, the CNN includes a semantic segmentation CNN configured to image pixels into one or more classes (e.g., lane marker classes). Each of the one or more classes corresponds to semantic content. In some embodiments, the data processing model **250** includes an efficient residual factorized ConvNet (ERFNet) for real-time semantic segmentation.

[0088] In some embodiments, a recurrent neural network (RNN) is applied in the vehicle data processing model **250** to process vehicle data **112**. Nodes in successive layers of the RNN follow a temporal sequence, such that the RNN exhibits a temporal dynamic behavior. In an example, each node **520** of the RNN has a time-varying real-valued activation. It is noted that in some embodiments, two or more types of vehicle data are processed by the data processing module **228**, and two or more types of neural networks (e.g., both CNN and RNN) are applied in the same vehicle data processing model **250** to process the vehicle data jointly.

[0089] The training process is a process for calibrating all of the weights w_i for each layer of the neural network **500** using training data **248** which is provided in the input layer **502**. The training

process typically includes two steps, forward propagation and backward propagation, which are repeated multiple times until a predefined convergence condition is satisfied. In the forward propagation, the set of weights for different layers are applied to the input data and intermediate results from the previous layers. In the backward propagation, a margin of error of the output (e.g., a loss function) is measured, e.g., in a loss control module **412** in FIG. **4**, and the weights are adjusted accordingly to decrease the error. The activation function **532** is optionally linear, rectified linear unit, sigmoid, hyperbolic tangent, or of other types. In some embodiments, a network bias term **b** is added (**534**) to the sum of the weighted outputs from the previous layer before the activation function **532** is applied. The network bias **b** provides a perturbation that helps neural network **500** avoid over fitting the training data. In some embodiments, the result of the training includes the network bias parameter **b** for each layer.

[0090] FIG. **6A** is a flow diagram of an exemplary process of classifying an image **602** on both a pixel level and an image level, in accordance with some embodiments. Operations described in process **600** are performed by a computer system (e.g., a computer system located on vehicles **102**, one or more servers **104**, one or more client devices **108**, or a combination thereof). The computer system includes one or more processors (e.g., CPU(s) **202** or CPU(s) **302**) and memory (e.g., memory **206** or memory **306**). The computer system optionally includes an event miner configured to detect whether a road image is an ambiguous image for lane marker classification. Lane markers (or lane markings) convey information to drivers by indicating which part of the road to use, what conditions lie ahead (e.g., merger of two lanes into one), and/or whether passing is allowed. Lane detection (or lane marker detection) refers to the task of detecting lane markers on the road. In some embodiments, lane marker detection and/or identification includes a multi-class segmentation task, wherein given an image (e.g., a road image), the computer system determines (e.g., predicts) that each pixel of the image belongs to a respective lane marker class (also called lane marker type).

[0091] In some embodiments, the computer system obtains the road image **602**, which includes a plurality of pixels. In some embodiments, the road image **602** is an image captured by one or more cameras (e.g., cameras **266** in FIG. **2**) integrated in or attached to a vehicle **102**. In some embodiments, a road image **602** includes one or more lane markers each of which has a respective lane marker class. For example, referring to FIG. **6A**, the road image **602** includes curbs **604** (e.g., a left curb **604-1** and a right curb **604-2** on the road image **602**). As used herein, a curb refers to a segment of the road where a road surface ends in a lateral direction that is generally perpendicular to a driving direction of the vehicle **102**. In some embodiments, a curb is treated as a lane marker class. The road image **602** further includes one or more dashed lane markers **606** (e.g., dashed lane marker **606-1** and dashed lane marker **606-2**), and one or more solid lane markers **608** (e.g., a center marker that separates lanes of two opposite directions, an edge marker that separates a drive lane and a shoulder area). The road image **602** further includes objects that belong to a no lane marker class **610**, such as the mountain **610-1**, road surface(s) **610-2**, and buildings **610-3**. In some embodiments, everything in the road image **602** except the curbs **604**, the dashed lane markers **606**, and the solid lane markers **608** is treated as having the no lane marker class **610**, and the road image **602** includes four lane marker classes including the curb mark **604**, dashed lane marker **606**, solid lane marker **608**, and no lane marker **610** classes.

[0092] In some embodiments, the computer device applies a classification neural network to classify each pixel of the plurality of pixels (or a group of pixels) of the road image to a respective road marker class (e.g., including, for each pixel, determining a plurality of probabilities associated with a plurality of lane marker classes). The classification neural network optionally includes a CNN (e.g., a fully convolutional network (FCN)). Optionally, the CNN has a kernel size of 3×3 . In an example, the CNN of the classification neural network has an effective receptive field (ERF). In some embodiments, the classification neural network includes one of a U-Net (i.e., a symmetric encoder-decoder network), a pyramid scene parsing network, and a multi-path refinement network.

[0093] In some embodiments, for each pixel of the plurality of pixels of an image, the computer system determines a plurality of probabilities **612** of classifying the respective pixel to a plurality of corresponding road marker classes **604-610**. For example, four probabilities **604P** (P1), **606P** (P2), **608P** (P3), and **610P** (P4) are determined for each pixel of the road image **602** to classify the respective pixel to a curb marker class **604**, a dashed lane marker class **606**, a solid lane marker class **608**, and a no lane marker class **610**, respectively. In some embodiments, the sum of the plurality of probabilities **612** for each pixel is equal to one or 100%. The curb marker class **604**, dashed lane marker class **606**, and solid lane marker class **608** form a subset of lane-based classes **614** in the plurality of marker classes **604-610**.

[0094] In some embodiments, the computer system applies hard thresholding **616** to classifying each pixel of the road image **602** based on the plurality of probabilities **612** of the respective pixel. In accordance with a determination that each of a subset of probabilities of a first pixel of the road image **602** is lower than respective hard threshold values $P_{sub.iTH}$ corresponding to the subset of lane-based classes **614**, the computer system does not associate (**618A**) the first pixel with any of the subset of lane-based classes, and classifies the first pixel to the no lane marker class **610**. Examples of respective hard threshold values $P_{sub.iTH}$ for the curb mark **604**, dashed lane marker **606**, solid lane marker **608**, and no lane marker **610** are 0.7, 0.75, 0.6, or 0.8, respectively. The respective hard threshold values $P_{sub.iTH}$ are optionally equal or different for the curb mark **604**, dashed lane marker **606**, solid lane marker **608**, and no lane marker **610**.

[0095] In an example, the plurality of probabilities **612** of classifying the first pixel to the curb mark **604**, dashed lane marker **606**, solid lane marker **608**, and no lane marker **610** are determined to be 0.05, 0.05, 0.6, and 0.3, respectively. And, in this example, the respective hard threshold values $P_{sub.iTH}$ are equal to 0.7. The highest probability $P_{sub.H}(0.6)$ corresponds to the solid lane marker **608** and is less than the threshold value of 0.7. The probabilities **604P** and **606P** corresponding to the curb mark **604** and dashed lane marker **606** are also less than threshold value of **0.7**. The computer system therefore classifies the first pixel to the “no lane marker” class **610**.

[0096] Conversely, in some embodiments, in accordance with a determination that one of a subset of probabilities **604P-608P** of a second pixel of the road image **602** is higher than a corresponding hard threshold value $P_{sub.iTH}$ of the subset of lane-based classes **604-608**, the computer system classifies (**618B**) the second pixel to a respective one of the subset of the lane-based classes **604-608**. For example, the plurality of probabilities **612** of classifying the second pixel to the curb mark **604**, dashed lane marker **606**, solid lane marker **608**, and no lane marker **610** are determined to be 0.1, 0.75, 0.05, and 0.1, respectively. The respective hard threshold values $P_{sub.iTH}$ are equal to 0.7. In accordance with a determination that the highest probability $P_{sub.H}$ (e.g., 0.75) corresponds to the dashed lane marker **606** and exceeds the respective hard threshold value $P_{sub.iTH}$, the computer system classifies the second pixel to the “dashed lane marker” class **606**.

[0097] In some embodiments, each hard threshold value $P_{sub.iTH}$ is predefined by the computer system (e.g., which optionally receives an input for the hard threshold value $P_{sub.iTH}$ manually entered by an operator of the computer system). In some embodiments, each hard threshold value $P_{sub.iTH}$ is determined in such a way as to obtain the best possible metric on a benchmark dataset. For each lane marker class, a higher threshold value $P_{sub.iTH}$ leads to fewer false positives (e.g., a determination that there is a lane marker when no lane marker is present) whereas a lower threshold value $P_{sub.iTH}$ produces more false positives. In some embodiments, after the autonomous driving model **252** has been trained, the model is benchmarked against a test set, and each hard threshold value $P_{sub.iTH}$ can be adjusted (e.g., varied) according to a first number of false positives and a second number of false negatives produced by the autonomous driving model **252**.

[0098] Referring to FIG. **6A**, in some embodiments, the computer system is configured to determine (**630**) whether a road image **602** is an ambiguous image for lane marker classification. Specifically, the computer system initializes an ambiguous lane pixel count ALP and a total lane pixel count TLP to zero for the road image **602**. For each pixel in the road image **602**, the computer

system identifies a class having a highest probability **612H** (P.sub.H) (e.g., using an arg max operation) and determines (**620**) whether the highest probability **612H** (P.sub.H) is among the probabilities P1, P2, and P3 corresponding to the subset of lane-based classes **614**, i.e., whether the identified class belongs to a predefined subset of lane-based classes **614** (e.g., including the dashed lane marker class **606**, the solid lane marker class **608**, and the curb marker class **610**). If the identified class belongs to the predefined subset of lane-based classes **614**, the computer system increments the total lane pixel count TLP by 1. Additionally, if the identified class belongs to one of the predefined subset of lane-based classes **614**, the computer system determines (**622**) whether the highest probability **612H** (P.sub.H) of the respective pixel lies in a range RP.sub.TH around the respective hard threshold value P.sub.iTH (e.g., $P_{sub.iTH} - \Delta < P_{sub.H} < P_{sub.iTH} + \Delta$), where Δ is a predetermined value defining the range, and examples of Δ include, but are not limited, to 0.02, 0.05, 0.1). The range RP.sub.TH is optionally symmetric or asymmetric with respect to the respective hard threshold value P.sub.iTH, and the value Δ is optionally equal or different for each of the subset of lane-based classes **614**. In accordance with a determination that the highest probability P.sub.H of the respective pixel lies in the range RP.sub.TH around the respective hard threshold value P.sub.iTH, the computer system increments the ambiguous lane pixel count ALP (e.g., by one). By these means, the computer system determines the total lane pixel count TLP and ambiguous lane pixel count ALP based on the probabilities **612** of the plurality of road marker classes of each pixel.

[0099] The computer system further determines a fraction (e.g., ratio) F between the ambiguous lane pixel count ALP and the total lane pixel count TLP and compares (**624**) the fraction F with a threshold fraction F.sub.TH (also called threshold ratio). Examples of the threshold fraction F.sub.TH include, but are not limited to, 0.2, 0.25, and 0.3. Based on a comparison result, the computer system determines (**626**) whether the road image **602** is an ambiguous image for lane marker classification.

[0100] FIG. **6B** is a flow diagram of an exemplary process **650** of processing an ambiguous image (e.g., a road image **602**), in accordance with some embodiments. Operations described in the process **600** are performed by a computer system (e.g., a computer system located on vehicles **102**, one or more servers **104**, one or more client devices **108**, or a combination thereof). The computer system includes one or more processors (e.g., CPU(s) **202** or CPU(s) **302**) and memory (e.g., memory **206** or memory **306**). The computer system optionally includes an event miner configured to detect (**630**) whether a road image is an ambiguous image for lane marker classification.

[0101] In some embodiments, the computer system generates (**638**) an event in accordance with the determination that the road image **602** is an ambiguous image for lane marker classification (step **636**), e.g., based on a ratio between the ambiguous lane pixel count ALP and total lane pixel count TLP as described above for FIG. **6A**. In an example, the computer system adds a tag to the road image **602**, to indicate that the road image **602** is an ambiguous image for lane marker classification. In some embodiments, the road image **602** includes an image frame of a video stream (e.g., live stream) captured by a vehicle **102**. The computer system sends a request to the vehicle **102** to record and save the image **602** in local memory. Alternatively, in some embodiments, in accordance with a determination that the road image **602** is not (**632**) an ambiguous image, e.g., based on a ratio between the ambiguous lane pixel count ALP and total lane pixel count TLP, the process **650** ends (**634**), and the computer system does not generate an event.

[0102] In some embodiments, the road image **602** is then labeled (**640**). In some embodiments, the labeled image includes information (e.g., metadata) identifying one or more lane markers (e.g., locations of the lane markers) in the respective labeled image. In some embodiments, the labeled image includes information identifying one or more lanes in the respective labeled image (e.g., based on the locations of the one or more lane markers). Further, in some embodiments, the computer system adds (**642**) the labeled image to the training data **248** (FIGS. **2** and **3**) for training and retraining (**644**) the autonomous driving model **252** (FIG. **2**). In an example, the autonomous

driving model **252** (FIG. 2) is applied (e.g., locally) to predict a road condition **646**, allowing the vehicle **102** to be adjusted, controlled, or driven (**648**) adaptively and/or at least partially autonomously. The information identifying the one or more lanes is optionally applied as ground truth for training or retraining the autonomous driving model **252** (FIG. 2).

[0103] FIGS. 7A to 7F are exemplary images (e.g., road images) **710**, **720**, **730**, **740**, **750**, and **760** that have been identified as ambiguous images for lane marker classification, in accordance with some embodiments. Referring to FIG. 7A, a plurality of vehicles **102** are driving on a road. The road includes lanes that are defined by dashed lane markers **712**. The image **710** shows that the road has longitudinal cracks **714** and transverse cracks **716**. The longitudinal cracks **714** extend in parallel with a direction of travel of the vehicles **102**, and the transverse cracks **716** extend cross the road, i.e., in a direction that is substantially perpendicular to the direction of travel). The image **710** also shows that a surface of the road has deteriorated and appears differently at an area **718**. Referring to FIG. 7B, the image **720** includes two sets of lane markers **722** and **724** that have been clearly marked on a road with white paint. However, additional lane markers **726** and **728** are also observed on the road. These additional lane markers **726** and **728** correspond to traces left on the road after previous markers were not properly removed by a road crew. Referring to FIG. 7C, an image **730** of a road includes lane markers **732** that have faded or worn out, e.g., due to wear and tear over time. Referring to FIG. 7D, an image **740** of a road includes a set of lane markers **742** and a region **744** in which lane markers extended from the lane markers **742** are missing. Referring to FIG. 7E, an image **750** of a road **752** captured by one or more cameras **266** of a vehicle **102** at night. Due to lighting conditions or wear and tear of the road **752**, lane markers of the road **752** are not visible on the image **750**, regardless of whether these lane markers are present or not. Referring to FIG. 7F, the image **760** includes a road **762** having a series of lane markers **764** extending in parallel with a direction of travel of a vehicle **102**. The lane markers **764** intersect with a region **768** where lane markers are missing due to destruction (e.g., cracks **766**) on a corresponding road surface.

[0104] When road images captured by cameras **266** of a vehicle **102** are applied for model training or vehicle control, situations can occur where road markers on the road images are not recognized for different reasons (e.g., bad road conditions, poor image quality, poor ambient light conditions). Some road markers may be missing on the images. Some road markers appear on a drivable area by mistake. Some road markers are not sufficiently clear and cannot be recognized by the data processing model **250** (FIG. 2). Some road markers are not clear (e.g., have noise) and cannot be recognized by the data processing model **250**. As explained above, the computer system determines (**630**) whether each image in FIGS. 7A-7F is an ambiguous image for lane marker classification based on a plurality of probability **612** of each pixel corresponding to a plurality of lane marker classes **606-610**. By these means, the computer system monitors a level of missing, erroneous, or noisy lane marker features, and train or apply the data processing model **250** adaptively based on the monitored level.

[0105] FIGS. 8A to 8C are a flow diagram of a method **800** for lane marker (e.g., road marker) detection for driving a vehicle, in accordance with some embodiments. The method **800** is performed at a computer system that has one or more processors (e.g., CPU(s) **202** or CPU(s) **302**) and memory (e.g., memory **206** or memory **306**) (FIGS. 2 and 3). The memory stores one or more programs configured for execution by the one or more processors. In some embodiments, the operations shown in FIGS. 8A-8C correspond to instructions stored in the memory or other non-transitory computer-readable storage medium. The computer-readable storage medium may include a magnetic or optical disk storage device, solid state storage devices such as Flash memory, or other non-volatile memory device or devices. The instructions stored on the computer-readable storage medium may include one or more of: source code, assembly language code, object code, or other instruction format that is interpreted by one or more processors. Some operations in the method **800** may be combined and/or the order of some operations may be changed.

[0106] The computer system obtains (804) a plurality of road images captured by one or more cameras attached to the one or more vehicles 102. The one or more vehicles 102 execute (805) a model (that facilitates driving of the one or more vehicles 102). For each road image of the plurality of road images (e.g., road image 602 in FIG. 6A), the computer system determines (806) a fraction (e.g., ratio) of pixels having an ambiguous lane marker classification in the road image and determines (822) whether the road image is an ambiguous image for lane marker classification based on the fraction of pixels. For each road image, in accordance with a determination that the road image is an ambiguous image for lane marker classification, the computer system enables (842) labeling of the road image and adds (842) the labeled image into a corpus of training images for retraining the model.

[0107] In some embodiments, the computer system trains (802) the model using machine learning. The model facilitates driving of one or more vehicles 102. In some embodiments, the model includes a vehicle data processing model 250 (e.g., an autonomous driving model 252). Each vehicle data processing model 250 is trained to process corresponding vehicle data to implement a respective one of a plurality of vehicle tasks including, but not limited to, perception and object analysis 230, vehicle localization and environment mapping 232, vehicle drive control 234, vehicle drive planning 236, local operation monitoring 238, and vehicle driving behavior monitoring 240 (FIG. 2).

[0108] Referring to FIG. 8A, in some embodiments, for each road image of the plurality of road images, the respective road image includes (808) a plurality of pixels. Determining the fraction of pixels having an ambiguous lane marker classification includes classifying (810) each pixel of the plurality of pixels into a class (e.g., lane marker class) selected from a group consisting of: a dashed lane marker class 606, a solid lane marker class 608, a curb marker class 604, and a no lane marker class 610. As used herein, a lane includes the road surface between the lane markers (or lane lines). A curb marker class 604 is a class of lane that depicts where a road surface ends. In some embodiments, the “no lane marker” class 610 includes everything in the image except a solid lane marker 608, a dashed lane marker 606, and a curb 604. For example, a captured image can include the sky, or surrounding trees. Image pixels corresponding to these objects have the “no lane marker” classification.

[0109] In some embodiments, the computer device applies a classification neural network to classify each pixel of the plurality of pixels of the road image to a respective road marker class (e.g., including, for each pixel, determining a plurality of probabilities associated with a plurality of lane marker classes). The classification neural network optionally includes a CNN (e.g., a fully convolutional network (FCN)). Optionally, the CNN has a kernel size of 3×3 . In an example, the CNN of the classification neural network has an effective receptive field (ERF). In some embodiments, the classification neural network includes one of a U-Net (i.e., a symmetric encoder-decoder network), a pyramid scene parsing network, and a multi-path refinement network.

[0110] In some embodiments, the classifying includes, for each road image of the plurality of road images, the computer system determines (812), for each pixel of the plurality of pixels, probabilities 612 (e.g., confidences or likelihoods) that the pixel should be classified into the dashed lane marker class 606, the solid lane marker class 608, the curb marker class 604, and the no lane marker class 610 (FIG. 6). In some embodiments, the respective probability has a value from 0 to 1, and the sum of the probabilities for each pixel equals to 1.

[0111] Referring to FIG. 8A, in some embodiments, the computer system identifies (814), for each pixel of the plurality of pixels of a road image, a highest probability PH from the probabilities 612. In some embodiments, in accordance with a determination that the highest probability $P_{sub.H}$ is less than a threshold value (e.g., $P_{sub.iTH}$ in FIG. 6A) (e.g., 0.7, 0.75, or 0.8), the computer system classifies (816) the pixel into the no lane marker class 610. For example, the plurality of pixels include a pixel whose probabilities are $P(\text{no lane marker})=0.2$, $P(\text{solid lane marker})=0.1$, $P(\text{dashed lane marker})=0.6$, and $P(\text{curb})=0.1$, and the threshold value is 0.7. In this example, even

though the dashed lane marker class has the highest probability, the computer system assigns the pixel as belonging to the “no lane marker” class because its probability value (0.6) is less than the threshold value of 0.7.

[0112] In some embodiments, in accordance with a determination that the highest probability $P_{\text{sub.H}}$ is greater (e.g., or at least equal to) than a threshold value (e.g., 0.7, 0.75, or 0.8), the computer system determines (818) that the highest probability $P_{\text{sub.H}}$ corresponds to an assigned lane marker class, and classifies (820) the pixel into the assigned lane marker class. For example, the plurality of pixels include a pixel whose probabilities are $P(\text{no lane marker})=0.1$, $P(\text{solid lane marker})=0.8$, $P(\text{dashed lane marker})=0.05$, and $P(\text{curb})=0.05$, and the threshold value is 0.7. In this example, the solid lane marker class 608 has the highest probability $P_{\text{sub.H}}$. The computer system assigns the pixel as belonging to the “solid lane marker” class 608 because its probability value (0.8) is greater (or at least equal to) the threshold value of 0.7.

[0113] With continued reference to FIG. 8B, the computer system determines (822), for each road image of the plurality of road images, based on the fraction of pixels, whether the road image is an ambiguous image for lane marker classification. In some embodiments, determining the fraction of pixels having an ambiguous lane marker classification includes initializing (824) (e.g., setting) a total lane pixel count TLP to zero. In some embodiments, determining the fraction of pixels having an ambiguous lane marker classification includes initializing (826) an ambiguous lane pixel count ALP to zero.

[0114] In some embodiments, the computer system determines (828), for each pixel of the plurality of pixels in the respective road image, a class that has a highest probability $P_{\text{sub.H}}$.

[0115] In some embodiments, in accordance with a determination that the class that has the highest probability ($P_{\text{sub.H}}$) belongs (830) to a predefined subset of the classes, the computer system increments a total lane pixel count TLP (e.g., by one).

[0116] In some embodiments, the predefined subset of classes includes (832) the dashed lane marker class, the solid lane marker class, and the curb marker class.

[0117] In some embodiments, in accordance with a determination that the highest probability is within a predetermined range of values from a first threshold value (e.g., 0.7, 0.75, or 0.8), the computer system increments (834) the ambiguous lane pixel count ALP (e.g., by one). For example, the predetermined range of values can be defined by [first threshold value minus delta, first threshold value plus delta], where delta is, e.g., 0.05 or 0.1.

[0118] In some embodiments, for each pixel of the plurality of pixels in the respective road image, in accordance with a determination that the class that has the highest probability belongs to a predefined subset of the classes, the computer system ignores (836) the pixel if the highest probability does not meet the first threshold value (e.g., 0.7, 0.75, or 0.8).

[0119] In some embodiments, the computer system determines (838) a ratio (e.g., fraction) of the ambiguous lane pixel count ALP to the total lane pixel count TLP.

[0120] In some embodiments, in accordance with a determination that the ratio is above (or at least equal to) a second threshold value (e.g., 0.2 or 0.3), the computer system determines (840) that the road image is an ambiguous image for lane marker classification.

[0121] With continued reference to FIG. 8C, the method 800 includes, for each road image of the plurality of road images, in accordance with a determination that the road image is an ambiguous image for lane marker classification, the image is labeled (842) (e.g., by a human). The computer system adds the labeled image into a corpus of training images for re/training the model.

[0122] In some embodiments, the image is an ambiguous image for lane marker classification when the image does not clearly depict that there is no lane marker. In some embodiments, an image is an ambiguous image for lane marker classification when the image does not clearly depict that there are lane markers. This is illustrated in FIGS. 7A to 7F.

[0123] In some embodiments, the image is an ambiguous image when the image does not clearly depict where the lane lines are, or what type of marker/line (e.g., a lane marker, such as dashed,

solid, curb line, certain color lane marker, etc.) the lane line is. This is illustrated in FIGS. 7A to 7F. [0124] In some embodiments, enabling labeling of the image includes receiving (844) user input for the labeling.

[0125] In some embodiments, the computer system retrain the model using (846) machine learning.

[0126] In some embodiments, the model includes (848) an autonomous driving model (e.g., autonomous driving model 252). Retraining the model includes modifying the autonomous driving model based at least in part on the added labeled images (e.g., and the ambiguous image) such that the one or more vehicles 102 can be driven at least partially autonomously using the autonomous driving model.

[0127] In some embodiments, each labeled image includes (850) information identifying one or more lane markers (e.g., locations of the lane markers) in the respective labeled image; and/or information identifying one or more lanes in the respective labeled image (e.g., based on the locations of the one or more lane markers).

[0128] In some embodiments, the information identifying one or more lane markers includes ground truth of the ambiguous image, manually marked on the ambiguous image, so that the model better recognizes lane markers in future images that would otherwise be ambiguous images.

[0129] In some embodiments, adding the labeled image, corresponding to a respective road image, into the corpus of training images for retraining the model includes transmitting (852) the labeled image to a remote server to be added to the corpus of training images.

[0130] In some embodiments, after the retraining, the computer system distributes (854) the modified autonomous driving model to the one or more vehicles 102, including a first vehicle. The modified autonomous driving model is configured to process road images captured by the first vehicle to facilitate at least partially autonomously driving the first vehicle.

[0131] In some embodiments, the road images are captured by the first vehicle in real time.

[0132] In some embodiments, the road images are captured by the first vehicle automatically and without user intervention.

[0133] In some embodiments, the autonomous driving model is configured to modify an autonomous driving variable based on the road images captured by the first vehicle. In some embodiments, modifying an autonomous driving variable includes discarding (e.g., ignoring) the road images. In some embodiments, modifying an autonomous driving variable includes using another model, such as following a vehicle that is in front.

[0134] It should be understood that the particular order in which the operations in FIGS. 8A-8C have been described are merely exemplary and are not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to perform lane marker detection for vehicle driving as described herein. Additionally, it should be noted that details described above with respect to FIGS. 1-7 are also applicable in an analogous manner to the method 800 described above with respect to FIGS. 8A-8C. For brevity, these details are not repeated here.

[0135] In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media, or communication media including any medium that facilitates transfer of a computer program from one place to another (e.g., according to a communication protocol). In this manner, computer-readable media generally may correspond to (1) tangible computer-readable storage media which is non-transitory or (2) a communication medium, such as a signal or carrier wave. Data storage media may be any available media that can be accessed by one or more computers or one or more processors to retrieve instructions, code and/or data structures for implementation of

the embodiments described in the present application. A computer program product may include a computer-readable medium.

[0136] The terminology used in the description of the embodiments herein is for the purpose of describing particular embodiments only and is not intended to limit the scope of claims. As used in the description of the embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, elements, and/or components, but do not preclude the presence or addition of one or more other features, elements, components, and/or groups thereof.

[0137] It will also be understood that, although the terms first and second may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first vehicle could be termed a second vehicle, and, similarly, a second vehicle could be termed a first vehicle, without departing from the scope of the embodiments. The first vehicle and the second vehicle are both vehicles **102**, but they are not the same vehicle.

[0138] The description of the present application has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the invention in the form disclosed. Many modifications, variations, and alternative embodiments will be apparent to those of ordinary skill in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. The embodiments are described in order to best explain the principles of the invention, the practical application, and to enable others skilled in the art to understand the invention for various embodiments and to best utilize the underlying principles and various embodiments with various modifications as are suited to the particular use contemplated. Therefore, the scope of the claims is not to be limited to the specific examples of the embodiments disclosed. Modifications and other embodiments are intended to be included within the scope of the appended claims.

Claims

1. A method for lane marker detection for vehicle driving, comprising: at a computer system including one or more processors and memory: obtaining a road image captured by a vehicle; determining whether the road image is ambiguous for lane marker classification; and in accordance with a determination that the road image is ambiguous for lane marker classification: generating a labeled road image, including identifying one or more lane markers in the road image; adding the labeled road image to a corpus of training data for training a model to generate an autonomous driving model; and distributing the autonomous driving model to one or more vehicles, wherein the autonomous driving model is configured to process road images captured by the one or more vehicles to facilitate at least partially autonomously driving the one or more vehicles.
2. The method of claim 1, wherein determining whether the road image is ambiguous for lane marker classification includes: determining a plurality of regions of the road image; and classifying each region of the plurality of regions to have a lane marker classification from a plurality of lane marker classifications.
3. The method of claim 2, wherein the plurality of lane marker classifications include: a dashed lane marker class, a solid lane marker class, a curb marker class, and a no lane marker class.
4. The method of claim 2, wherein each region of the plurality of regions corresponds to one respective single pixel of the road image.
5. The method of claim 2, wherein each region of the plurality of regions corresponds to at least

two pixels of the road image.

6. The method of claim 2, wherein the classifying includes: determining, for each respective region of the plurality of regions, probabilities that the respective region should be classified into the plurality of lane marker classifications.

7. The method of claim 6, further comprising: for each region of the plurality of regions in the road image: determining a class that has a highest probability; in accordance with a determination that the class that has the highest probability belongs to a predefined subset of classes, incrementing a total lane pixel count; in accordance with a determination that the highest probability is within a predetermined range of values from a first threshold value, incrementing an ambiguous lane pixel count; and determining a ratio of the ambiguous lane pixel count to the total lane pixel count.

8. The method of claim 7, further comprising: in accordance with a determination that the ratio is above a second threshold value, determining that the road image is ambiguous for lane marker classification.

9. The method of claim 7, further comprising: for each region of the plurality of regions in the road image: in accordance with a determination that the class that has the highest probability belongs to the predefined subset of classes, ignoring the respective region if the highest probability does not meet the first threshold value.

10. The method of claim 1, the labeled road image includes: information identifying one or more lane markers in the labeled road image; and/or information identifying one or more lanes in the labeled road image.

11. A computer system, comprising: one or more processors; and memory storing one or more programs configured for execution by the one or more processors, the one or more programs comprising instructions for: obtaining a road image captured by a vehicle; determining whether the road image is ambiguous for lane marker classification; and in accordance with a determination that the road image is ambiguous for lane marker classification: generating a labeled road image from the road image, including identifying one or more lane markers in the road image; adding the labeled road image to a corpus of training data for training a model to generate an autonomous driving model; and distributing the autonomous driving model to one or more vehicles, wherein the autonomous driving model is configured to process road images captured by the one or more vehicles to facilitate at least partially autonomously driving the one or more vehicles.

12. The computer system of claim 11, wherein the one or more vehicles includes the vehicle that captures the road image.

13. The computer system of claim 11, wherein the instructions for adding the labeled road image to the corpus of training data include instructions for: transmitting the labeled road image to a remote server to be added to the corpus of training images.

14. The computer system of claim 11, wherein the instructions for determining whether the road image is ambiguous for lane marker classification include instructions for: determining a plurality of regions of the road image; and classifying each region of the plurality of regions to have a lane marker classification from a plurality of lane marker classifications.

15. The computer system of claim 14, wherein the instructions for the classifying include instructions for: determining, for each respective region of the plurality of regions, probabilities that the respective region should be classified into the plurality of lane marker classifications.

16. The computer system of claim 15, the one or more programs further comprising instructions for: determining, for each region of the plurality of regions, a highest probability from the probabilities; and in accordance with a determination that the highest probability is less than a threshold value, classifying the region into a no lane marker class.

17. The computer system of claim 15, the one or more programs further including instructions for: determining, for each region of the plurality of regions, a highest probability from the probabilities; and in accordance with a determination that the highest probability is greater than a threshold value: determining that the highest probability corresponds to an assigned class; and classifying the

respective region into the assigned class.

18. A non-transitory computer-readable storage medium storing one or more programs configured for execution by one or more processors of a computer system, the one or more programs comprising instructions for: obtaining a road image captured by a vehicle; determining whether the road image is ambiguous for lane marker classification; and in accordance with a determination that the road image is ambiguous for lane marker classification: generating a labeled road image from the road image, including identifying one or more lane markers in the road image; adding the labeled road image to a corpus of training data to train a model to generate an autonomous driving model; and distributing the autonomous driving model to one or more vehicles, wherein the autonomous driving model is configured to process road images captured by the one or more vehicles to facilitate at least partially autonomously driving the one or more vehicles.

19. The non-transitory computer-readable storage medium of claim 18, wherein the instructions for determining whether the road image is ambiguous for lane marker classification include instructions for: determining whether each region of a plurality of regions of the road image is ambiguous for lane marker classification.

20. The non-transitory computer-readable storage medium of claim 19, wherein the instructions for determining whether each region of the plurality of regions of the road image is ambiguous for lane marker classification include instructions for: determining a set of probabilities of classifying each region of a plurality of regions of the road image into a plurality of predefined classes, wherein whether each region of the plurality of regions of the road image has the ambiguous lane marker classification is determined based on values of the set of probabilities of the respective region.
